

Preempting Fermion Sign Problem: Unveiling Quantum Criticality through Nonequilibrium Dynamics in Imaginary Time

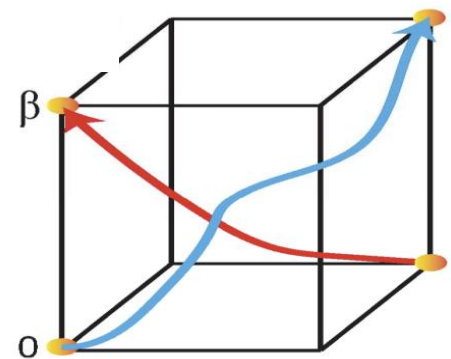
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The **fermion sign problem** is a major obstacle for simulating quantum many-body systems. To overcome this, we introduce a framework based on **imaginary-time nonequilibrium critical dynamics**. We show that critical properties can be accurately extracted from short-time relaxation, where the sign problem remains manageable for quantum Monte-Carlo (QMC). We apply this method to the sign-problematic **SU(3) Hubbard model** with Dirac fermions. Our results provide the first numerically exact quantum phase diagram, revealing a continuous transition between a Dirac semi-metal and an SU(3) antiferromagnetic phase. This transition uncovers an **unconventional Gross-Neveu universality class** that reshapes current understanding of Gross-Neveu criticality.

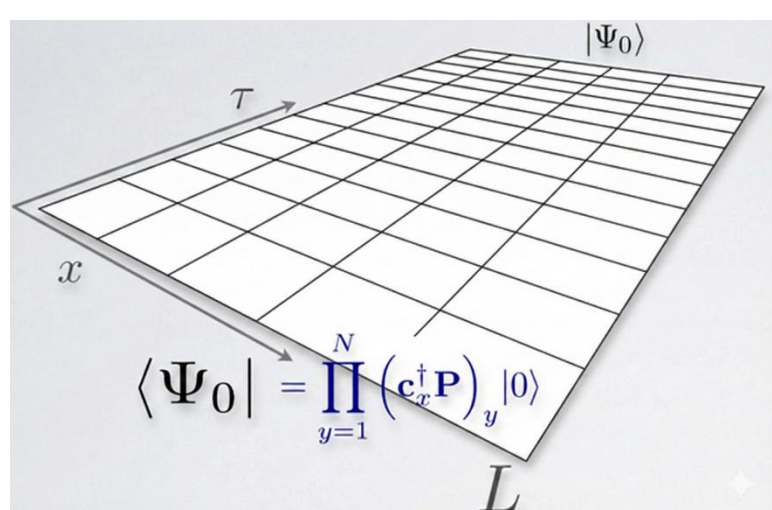
Fermion sign problem

➤ Fermion statistics: the **minus sign** in the wave function induced by exchanging fermions.



➤ Projective quantum Monte-Carlo: access ground-state property via **imaginary-time evolution**

Initial state: $|\psi_0\rangle$ → Ground state: $|\psi_G\rangle$



$$|\psi_G\rangle = e^{-\hat{H}\tau} |\psi_0\rangle$$

Monte-Carlo Sampling the partition:

$$Z = \sum_{[\phi]} \langle \psi_0 | e^{-\hat{H}[\phi]\tau} | \psi_0 \rangle$$

➤ Sign problem: non-positive weights!

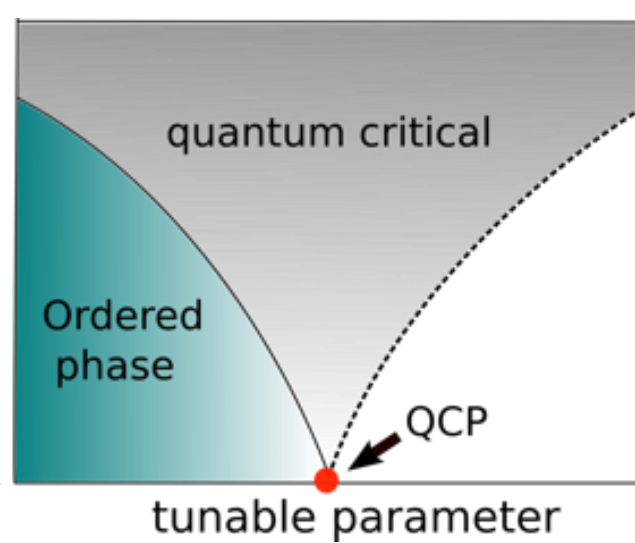
$$\text{Errors} \propto \frac{1}{\langle \text{sign} \rangle} \sim e^{N\tau\Delta f}$$

❑ Impact: **exponential** time complexity!

Universal short-time critical dynamics

➤ Quantum criticality:

- ❑ Diverging critical fluctuations at continuous phase transitions
- ❑ Requires $\tau \rightarrow \infty$ to ground state (**inaccessible** at sign-problematic systems)



➤ **Nonequilibrium** scaling theory:

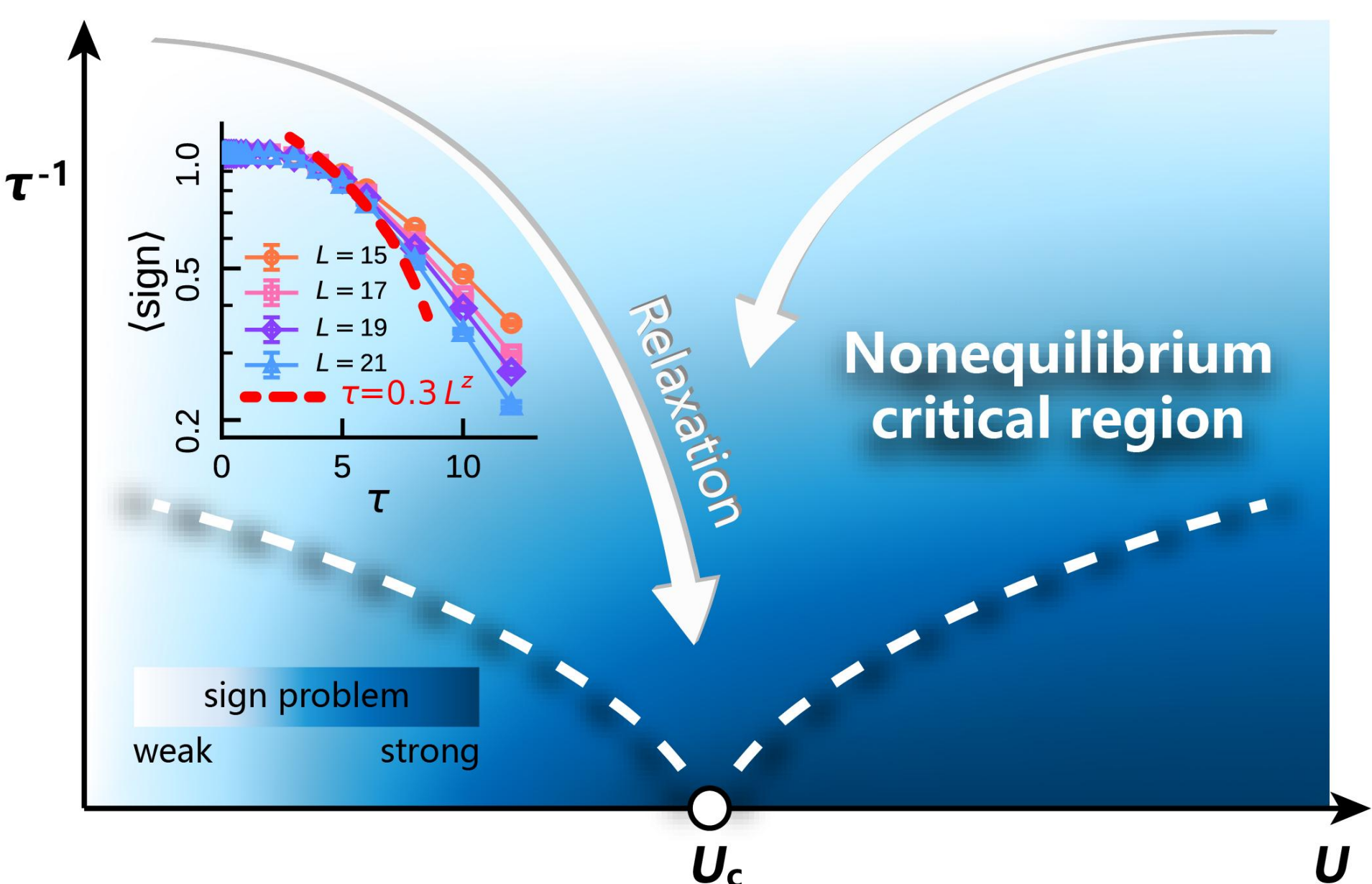
$$O(\tau, U, L) = L^{-\kappa} f_0((U - U_c)L^{1/\nu}, \tau L^{-z})$$

- ❑ L : system size | τ : imaginary time | κ : scaling dimension of O
- ❑ z : dynamical exponent ($z = 1$ for Dirac quantum criticality)
- ❑ Initial state $|\psi_0\rangle$: stable fix point

Initial state: $|\psi_0\rangle$ → Nonequilibrium state: $|\psi(\tau)\rangle$ → **Very short τ** (No longer required) Ground state: $|\psi_G\rangle$

❑ Critical properties and ground-state phase diagrams can be determined from **short-time** relaxation.

Outrunning the sign problem

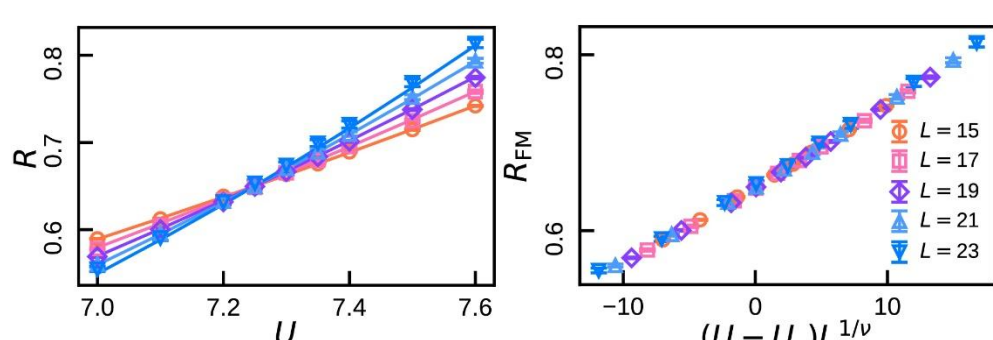


- Conventional: $\tau = 2 \sim 3 L^z$ → severe sign problem
- Our approach: $\tau = 0.3 L^z$ → **mild sign problem!** (exponential speedup in simulation)

➤ **Benchmark:** single-Dirac-fermion Hubbard model

$$H = \sum_p t_R c_{i\uparrow}^\dagger c_{i+R\uparrow} + \text{h.c.} + U \sum_i \left(n_{i\uparrow} - \frac{1}{2} \right) \left(n_{i\downarrow} - \frac{1}{2} \right)$$

- ❑ Square lattice, $t_R = \frac{i(-1)^{R_x}}{\pi \sin \frac{R_x}{L}} \delta_{R_y,0} + \frac{(-1)^{R_x}}{\pi \sin \frac{R_x}{L}} \delta_{R_y,0}$ for single Dirac cone
- ❑ Small U : Dirac semi-metal (DSM) phase; Large U : ferromagnetic (FM) phase



Take $\tau = 0.3 L^z$, correlation length ratio: $R_{FM} = f_R((U - U_c)L^{1/\nu})$

Take $U = U_c$, Structure factor: $S_{FM} = L^{-(1+\eta_\phi)} f_S(\tau L^{-z})$ Fermion correlation: $G_F = L^{-\eta_\psi} f_G(\tau L^{-z})$

Methods	U_c	ν^{-1}	η_ϕ	η_ψ
This work (from DSM)	7.220(37)	1.18(3)	0.36(3)	0.134(3)
This work (from FM)	7.214(44)	1.05(10)	0.35(3)	0.136(14)
Gutzwiller-PQMC	7.275(25)	1.19(3)	0.31(1)	0.136(5)
FRG	-	1.229	0.372	0.131

Application to SU(3) Hubbard model

➤ SU(N) fermion

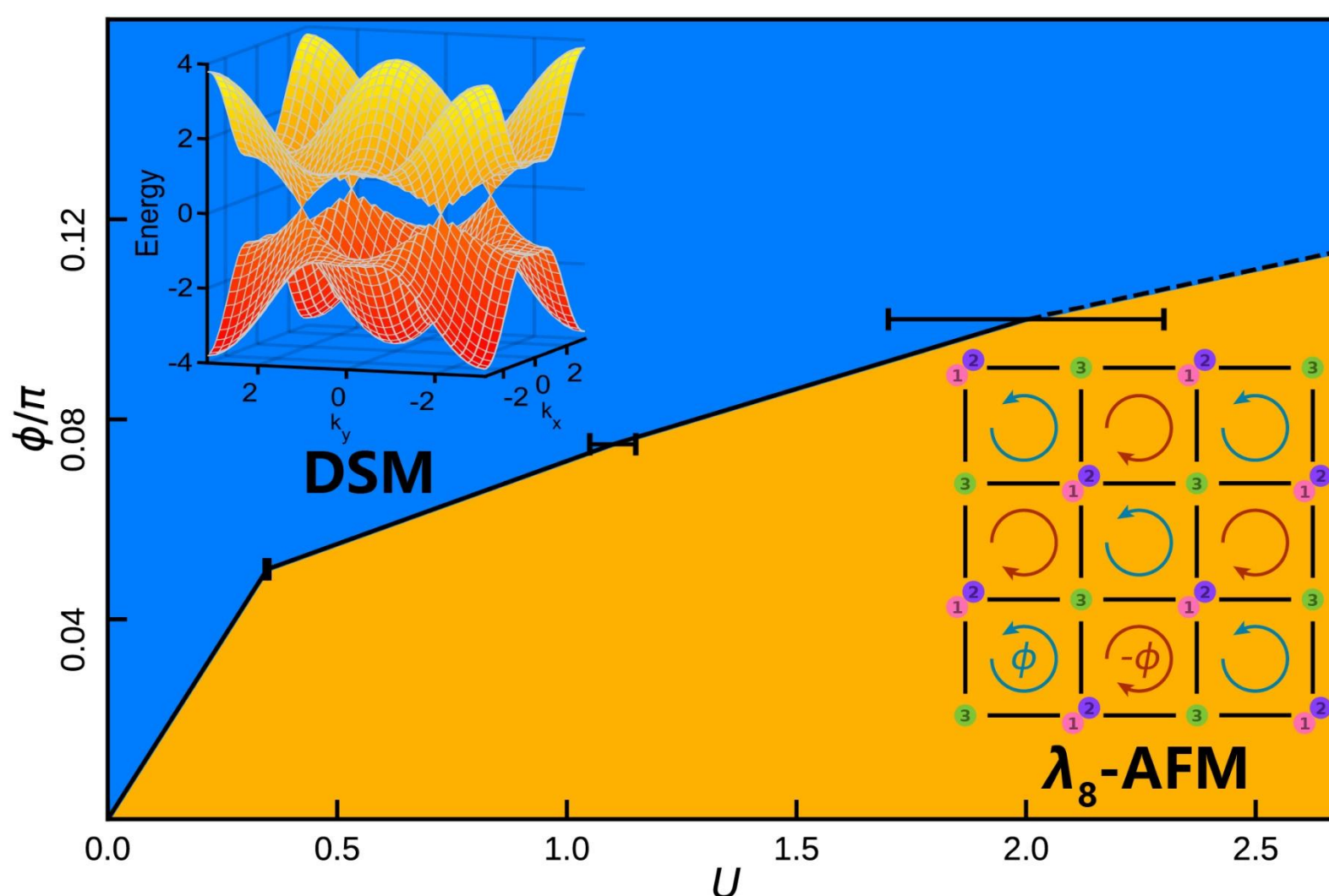
SU(2) → SU(N)

- ❑ Theoretical: possible exotic phases of matter, quarks...
- ❑ Experimental: Optical lattice of cold atoms, twisted 2d materials...
- ❑ Numerical: **sign problem arises for odd-N Hubbard interactions**

➤ SU(3) Hubbard model (minimal model with odd N)

$$H = \sum_{\langle ij \rangle \alpha} t_{ij} c_{i\alpha}^\dagger c_{j\alpha} + \frac{U}{2} \sum_i \left(\sum_\alpha n_{i\alpha} - \frac{3}{2} \right)^2$$

- ❑ With staggered flux ϕ : $t_{ij} = t e^{i\theta_{ij}}$, $\theta_{ij} = (-1)^{i_x+i_y}\phi/4$
- ❑ Symmetry: $SU(3) \times Z_2$ → $SU(3)$ on flavors, Z_2 on sublattices
- ❑ Phase diagram:



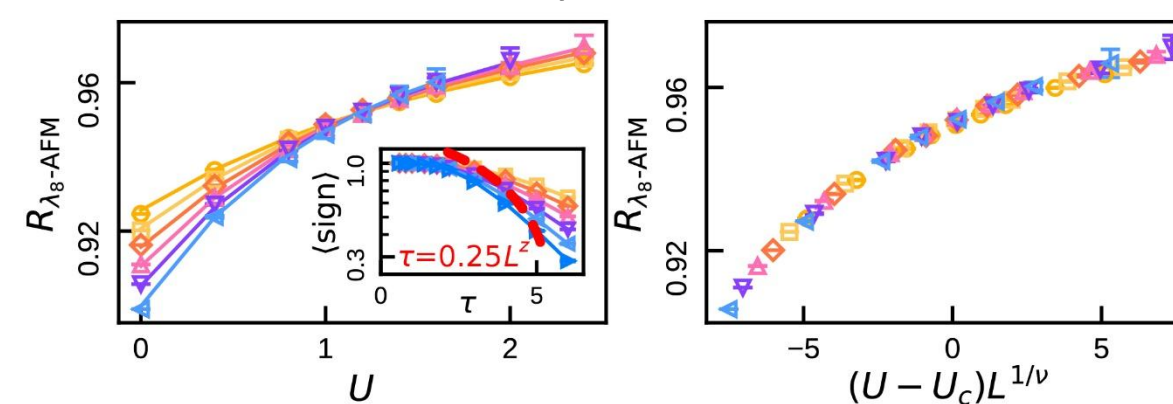
SU(3) generator Gell-Mann matrix λ_8

$$\begin{pmatrix} 1/\sqrt{3} & 0 & 0 \\ 0 & 1/\sqrt{3} & 0 \\ 0 & 0 & -2/\sqrt{3} \end{pmatrix}$$

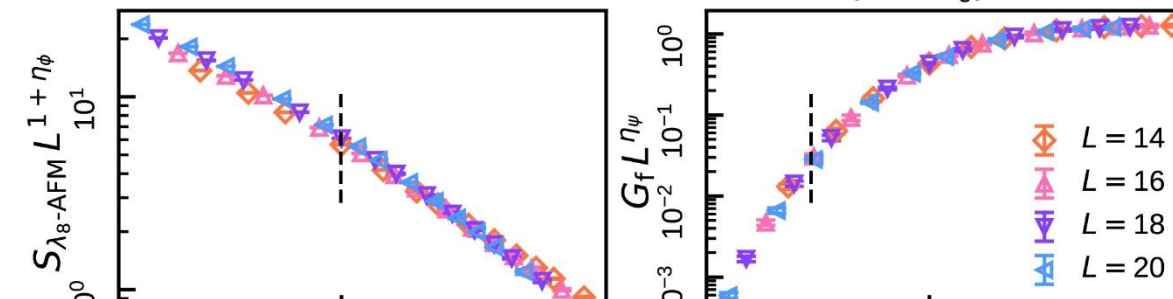
Staggered λ_8 order: flavor 3 on A sublattice flavor 1,2 on B sublattice

➤ Quantum criticality: short-time critical dynamics

- ❑ Take $\tau = 0.25 L^z$,
- ✓ Scaling of correlation length ratio $R_{\lambda_8\text{-AFM}} = f_R((U - U_c)L^{1/\nu})$

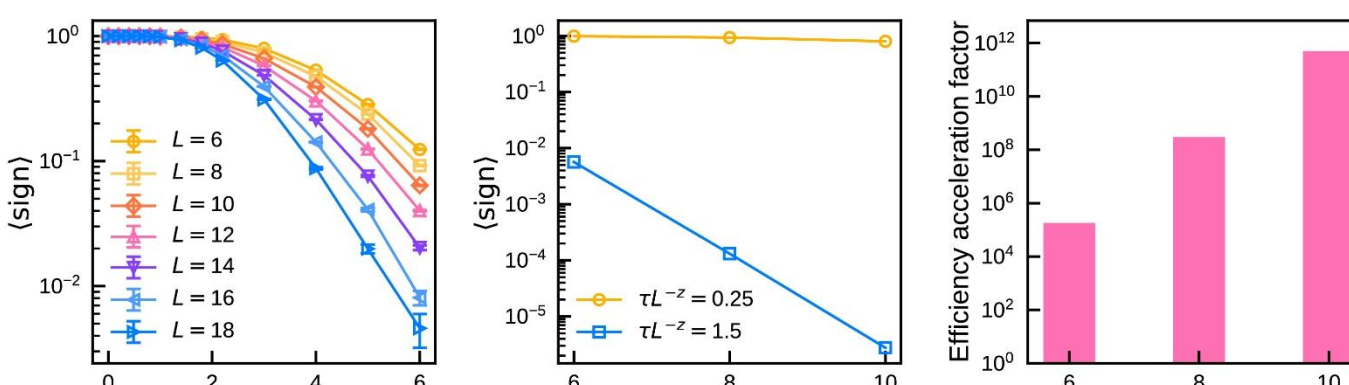


- ❑ Take $U = U_c$,
- ✓ Scaling of Structure factor: $S_{\lambda_8\text{-AFM}} = L^{-(1+\eta_\phi)} f_S(\tau L^{-z})$
- ✓ Scaling of Fermion correlation: $G_F = L^{-\eta_\psi} f_G(\tau L^{-z})$



✓ Critical exponents: $\nu^{-1} = 0.68(5)$, $\eta_\phi = 0.55$, $\eta_\psi = 0.15(3)$

➤ Sign problem behaviors



✓ **Billions-fold speedup in computation**

New universality class

This continuous phase transition in SU(3) Hubbard model defines a new universality class: **chiral SU(3)**

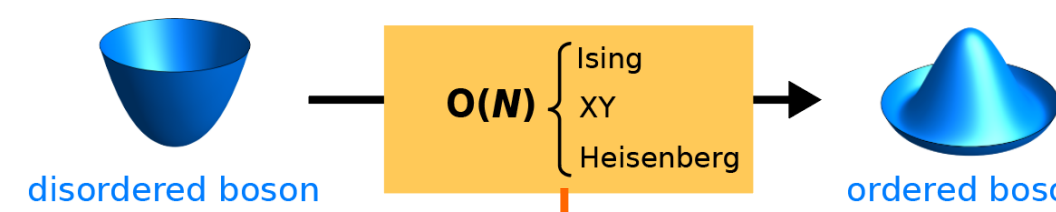
➤ Critical exponents comparison:

Universality class	ν^{-1}	η_ϕ	η_ψ
chiral SU(3) (this work)	0.68(5)	0.55(5)	0.15(3)
chiral Heisenberg	1.478	1.023	0.058
chiral XY	1.809	0.698	0.082
chiral Ising	0.750	0.865	0.011

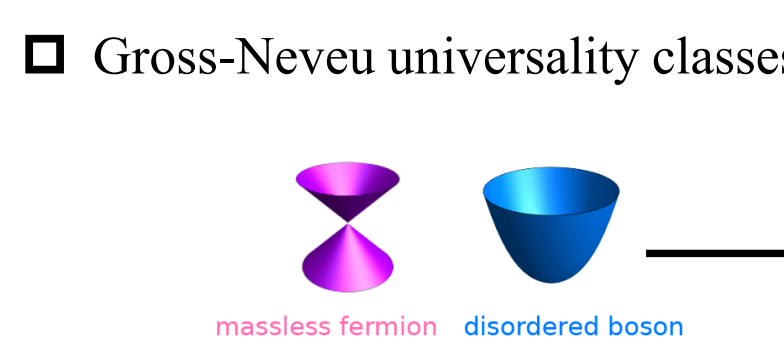
Distinct from all previously known Gross-Neveu universality classes ($d=2+1$, $N=6$)

➤ Unconventional Gross-Neveu transition

❑ O(N) universality classes



Historically, the Gross-Neveu universality classes have long been regarded as the chiral O(N_b) universality class.



chiral SU(3): with out O(N) counterpart

Presently, **our work extends the scope of Gross-Neveu transitions.**

Summary and outlook

➤ Summary:

- ❑ A powerfully generalizable framework: preempting the sign problem through nonequilibrium dynamics.
- ❑ First numerically exact solution of the SU(3) Hubbard model and revealing an unconventional chiral SU(3) universality class.

➤ Outlook:

- ❑ Solving other challenging QCPs, e.g. the metallic QCP...

References

1. Yin-Kai Yu, Zhi-Xuan Li, Shuai Yin*, Zi-Xiang Li*, Sci. Adv. 12, eadz4856 (2026)